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VIEWPOINT: PART OF A SPECIAL ISSUE ON MATCHING ROOTS TO THEIR ENVIRONMENT

Plasticity of rhizosphere hydraulic properties as a key for efficient utilization of scarce resources

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• **Background** It is known that the soil near roots, the so-called rhizosphere, has physical and chemical properties different from those of the bulk soil. Rhizosphere properties are the result of several processes: root and soil shrinking/swelling during drying/wetting cycles, soil compaction by root growth, mucilage exuded by root caps, interaction of mucilage with soil particles, mucilage shrinking/swelling and mucilage biodegradation. These processes may lead to variable rhizosphere properties, i.e. the presence of air-filled gaps between soil and roots; water repellence in the rhizosphere caused by drying of mucilage around the soil particles; or water accumulation in the rhizosphere due to the high water-holding capacity of mucilage. The resulting properties are not constant in time but they change as a function of soil condition, root growth rate and mucilage age.

• **Scope** We consider such a variability as an expression of rhizosphere plasticity, which may be a strategy for plants to control which part of the root system will have a facilitated access to water and which roots will be disconnected from the soil, for instance by air-filled gaps or by rhizosphere hydrophobicity. To describe such a dualism, we suggest classifying rhizosphere into two categories: class A refers to a rhizosphere covered with hydrated mucilage that optimally connects roots to soil and facilitates water uptake from dry soils. Class B refers to the case of air-filled gaps and/or hydrophobic rhizosphere, which isolate roots from the soil and may limit water uptake from the soil as well water loss to the soil. The main function of roots covered by class B will be long-distance transport of water.

• **Outlook** This concept has implications for soil and plant water relations at the plant scale. Root water uptake in dry conditions is expected to shift to regions covered with rhizosphere class A. On the other hand, hydraulic lift may be limited in regions covered with rhizosphere class B. New experimental methods need to be developed and applied to different plant species and soil types, in order to understand whether such dualism in rhizosphere properties is an important mechanism for efficient utilization of scarce resources and drought tolerance.

Key words: Rhizosphere, hydraulic properties, root water uptake, mucilage, root–soil contact, gaps.

INTRODUCTION

The function of roots in nutrient acquisition and exudate release is intimately linked to root water uptake and spatial distribution of water in soils. Despite numerous investigations ranging from molecular studies of the expression of aquaporins (Maurel *et al.*, 2008), to the measurements at the tissue level of root hydraulic conductivities (Steudle, 2001), up to field-scale models of soil and plant interactions (Jarvis, 2011), there is still great uncertainty in the prediction of root water uptake. It is still poorly understood which parts of the root system contribute the most to the total plant water uptake at varying soil water distributions.

The difficulty in predicting root water uptake depends on the short-term changes in membrane conductivity resulting from aquaporin expression, as well as on the changes in axial and radial conductivity of roots during root development, i.e. the formation of apoplastic barriers (casparian bands and suberin lamellae) in the endo- and exodermis. Zwieniecki *et al.* (2003) measured the axial and radial conductivity of segments of maize roots. They found that the root surface was not

equally involved in water uptake, and that, as the water potential dissipated along root xylem, the more distal segments of roots were less active in uptake. Root sealing is a strategy of plants to isolate the upper roots from soil and increase the water uptake in the more distal parts.

Root functions also depend on soil properties, which are highly variable in time and space. Soil hydraulic conductivity decreases by several orders of magnitude as soil water content decreases. In a drying soil, the soil hydraulic conductivity can become the major source of resistance in the soil–plant–atmosphere continuum, and a water depletion zone can develop around roots, further limiting the water availability to plants. Ultimately, individual roots may shrink and lose contact with the soil. The contribution of these roots to the overall water uptake may become severely limited. The relative importance of the soil and roots can be understood only by means of models that couple water flow in soil and roots and that include root–soil interactions and feedbacks, as discussed by Draye *et al.* (2010)

A physical model of water flow from soil to the surface of a single root was first formulated by Gardner (1960). Since then,

the model has been implemented in more complex three-dimensional root water uptake models. A common assumption of these models is that the hydraulic properties of soil do not vary as a function of distance from the roots. The models predict that as the soil dries, a water depletion zone develops around the active roots. Indeed, there is experimental evidence showing water depletion zones around roots (Macfall *et al.*, 1990).

However, other observations contradict the results of Gardner's model. Young (1995) reported higher water content in the soil region near roots of wheat than in the adjacent bulk soil. Recent experiments with neutron radiography (Carminati *et al.*, 2010) showed that the soil near the roots of lupins was wetter than the bulk soil during drying, but it remained dry after irrigation. The water content in the soil next to roots varied over time depending on the drying/wetting history of the soil.

Such observations of unexpected water contents can be explained if one accepts that the hydraulic properties of the soil near the roots differ from those of the adjacent bulk soil. Indeed, there is a large volume of literature about the specific and unique properties of the soil near roots, the so-called rhizosphere. Most of the literature agrees that the properties of the rhizosphere are physically, chemically and biologically different from those of the bulk soil (Hinsinger *et al.*, 2009). However, there is some confusion about what these specific properties actually are: for instance, does the rhizosphere hold a higher or lower water content than the bulk soil at a specific water potential? As discussed in the following sections, the contradictions of the properties of the rhizosphere reflect the dynamic and plastic nature of the processes occurring in the rhizosphere.

We propose to extend the concept of root plasticity to the rhizosphere. We will focus on the properties and role of mucilage exuded by roots. Mucilage is a very likely candidate to explain the observed dynamics of rhizosphere properties: (1) it has a very high water-holding capacity; (2) it may become hydrophobic once dried; and (3) it is primarily present along younger parts of the root system.

We will start with a synthesis of our recent results from X-ray computed tomography (CT) and neutron tomography/radiography. Based on these observations, we suggest assigning roots to two classes of rhizosphere hydraulic properties, i.e. class A (covered with hydrated mucilage); and class B (degraded mucilage/gaps). The distribution of the two classes is not constant in time, but changes with root growth and soil drying/wetting history. We discuss the extent of such a bimodal distribution of rhizosphere hydraulic properties and the consequences for root system water uptake and water redistribution (hydraulic lift) in the section 'Rhizosphere plasticity'. The extent of bimodal distribution strongly depends on existing knowledge concerning mucilage chemistry, production and degradation rate, as well as shrinking and swelling cycles, which will be addressed in the section 'Mucilage composition, distribution and function in the rhizosphere'.

We will discuss in detail factors related to the amount, quality, function and distribution of mucilage and any specific co-occurrence of mucilage and those factors which affect the extent of matric potential gradients around individual roots or spatial distribution of root hydraulic conductivity. Of

particular interest is the interaction between mucilage and soil particles and the consequences for soil hydraulic properties.

Finally, we will integrate the current knowledge about rhizosphere hydraulic properties and discuss the consequences for soil–plant water relations (see 'Rhizosphere and soil–plant water relations'). Knowledge gaps and research needs will be identified in the final section.

CLASSICAL MODEL OF WATER UPTAKE BY A SINGLE ROOT

First we introduce the classical models of water flow from soil to roots. Water flow to roots is driven by a water potential gradient between the bulk soil and the root surface. Let us first assume that a root is a cylinder of infinite length, the soil is a cylinder with a finite outer radius and concentric with the root, and neglect gravity. Typical profiles of water content θ and soil matric potential ψ towards a root are plotted in Fig. 1. The profiles were calculated solving the classical equation of water flow in soils (Richards' equation) under the steady rate assumption for a sandy soil and a constant water flux of 1 cm d^{-1} at the root surface. The soil is assumed to be a cylinder of radius 1 cm. The initial condition was a matric potential $\psi = -20 \text{ hPa}$. The solution was calculated until the matric potential at the root surface reached a value of $\psi = -1.5 \text{ MPa}$ – which is considered as the wilting point. The wilting point was reached after 2 d and 4 h. When the soil was wet ($t = 0$ –2 d), θ and ψ were almost flat. In other words, no significant water depletion was visible around the roots. The reason was that under wet conditions the soil hydraulic conductivity was much higher than the flow rate. However, as the soil became dry, a large gradient in ψ and a moderate gradient in θ developed in the first millimetres near roots ($t = 2 \text{ d}$ and 4 h). The gradients were caused by the radial geometry of the flow process, with the flux rate increasing towards the root, and by the non-linearity of the soil hydraulic conductivity, that dramatically decreased as the soil became dry. Under such conditions, the soil in the vicinity of the roots, the rhizosphere, may pose the main resistance to root water uptake. This flow process was first formulated by Gardner (1960) and since then it has been implemented in more complex three-dimensional root water uptake models (Tuzet *et al.*, 2003; Doussan *et al.*, 2006; Javaux *et al.*, 2008; Siqueira *et al.*, 2008). One of the main assumptions of the Gardner model is that the soil properties do not vary as a function of distance from the root. In other words, rhizosphere and bulk soil are assumed to have identical hydraulic properties. However, as reviewed in the following sections, rhizosphere and bulk soil may differ a lot.

AIR-FILLED GAPS BETWEEN SOIL AND ROOTS

Gardner's approach assumes uniform soil properties in the root zone. However, as plants transpire and soil becomes dry, roots shrink and may lose contact with the soil. Root shrinkage was already observed by Huck *et al.* (1970), who reported diurnal relative root shrinkage of 40 %. Root shrinkage may cause the formation of air-filled gaps between roots and soil. North and Nobel (1992) estimated that gaps pose the main resistance to

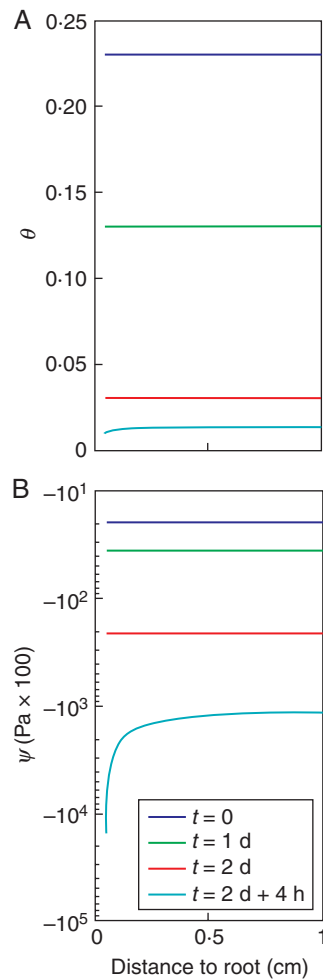


FIG. 1. Profiles of (A) water content θ and (B) matric potential ψ as a function of distance from a root assuming a constant uptake rate (1 cm d^{-1}). The profiles are calculated solving the Richards' equation in radial co-ordinates and assuming that the soil is a sand. During the first 2 d, the profiles were almost flat. As the soil became dry, a large drop in water potential developed in the soil near the root surface.

root water uptake at intermediate soil moisture. Passioura (1980) hypothesized that gaps could have been the cause of the reduced conductance of the root–soil system that he observed during a transpiration cycle (from low transpiration to high transpiration and back) when the soil was dry. However, further information about the spatial location of gaps and their relationship with soil water potential and transpiration rate is needed to draw final conclusions on the role of gaps.

We used X-ray CT to visualize the pore geometry in the vicinity of roots and the eventual formation of gaps between soil and roots (Carminati *et al.*, 2009). By using a spatial resolution of 0.1 mm, we were able to observe the formation of air-filled gaps around the tap root of lupins. Gaps were smaller for lateral roots. More recently, we studied the consequences of gaps on root water uptake (Carminati *et al.*, 2012). We grew four lupins (*Lupinus albus*) in cylinders of 8 cm in diameter and 20 cm in height filled with sandy soils. We measured

the transpiration rate, water content and soil water potential during a drying cycle and the subsequent irrigation. Simultaneously, we performed X-ray CT to observe the contact between soil and roots. The results showed that not all roots shrank to the same extent. Air gaps of 0.1–0.35 mm were observed for the tap root, while lateral roots shrank much less and seemed to remain in contact with the soil. Figure 2 (left) shows a horizontal cross-section of a lupin in a cylinder at a volumetric water content of $\theta = 0.03$. The cross-section corresponds to a soil depth of 5 cm and shows a continuous gap around the tap root and a better contact for a lateral branching from the tap root. Additionally, it was found that gaps occurred after the transpiration rate was severely reduced. The relationship between transpiration rate, soil matric potential and gap formation for four samples (Lupin I–IV) is illustrated in Fig. 2 (right). The relative transpiration on the y-axis is the actual transpiration divided by the maximum transpiration rate of the lupins when the soil was wet. The figure shows that gaps formed after the transpiration rate was already limited. From this figure, we conclude that gaps are a consequence rather than a cause of limited water availability. These experiments were performed with a voxel size of 0.1 mm. Gaps smaller than the voxel size would have therefore not been visible and it cannot be excluded that partial loss of contact between roots and soil occurred earlier. Recent advances in X-ray CT now allow scanning with voxel sizes of 10 μm (Mooney *et al.*, 2011), and will therefore answer this question. Investigations with other plant species and other soil types are also needed. Few attempts have been made to include information on partial loss of contact in models of root water uptake; in Veen *et al.* (1992) it was assumed that part of an individual root cylinder loses contact, and in Jensen *et al.* (1993) and Herkelrath *et al.* (1977) loss of contact was introduced by scaled soil water status. The possibility that some roots lose contact while others maintain contact has not been quantitatively addressed or mechanistically modelled.

WATER CONTENT DYNAMICS IN THE RHIZOSPHERE

What is the relationship between water content and water potential (the so-called water retention curve) in the rhizosphere? As illustrated in the modelling result of Fig. 1, large gradients in water potential develop in the first millimetres near roots as the soil dries. An alteration of the rhizosphere water retention curve would affect such gradients and the overall soil–plant water relations.

Experimental information on the soil hydraulic properties of the rhizosphere is limited and sometimes contradictory. For a detailed discussion, see 'Interaction between mucilage and soil particles' below and a recent review of rhizosphere biophysics given by Bengough (2012). The reason for the lack of knowledge on rhizosphere hydraulic properties is the difficulty in applying classical soil physics methods to the rhizosphere. Sample collection, as in Young (1995), revealed that the rhizosphere of wheat was wetter than the bulk soil. This result is in contradiction to studies showing a slight water repellence of the rhizosphere (Hallett *et al.*, 2003; Whalley *et al.*, 2005). Experiments with neutron radiography

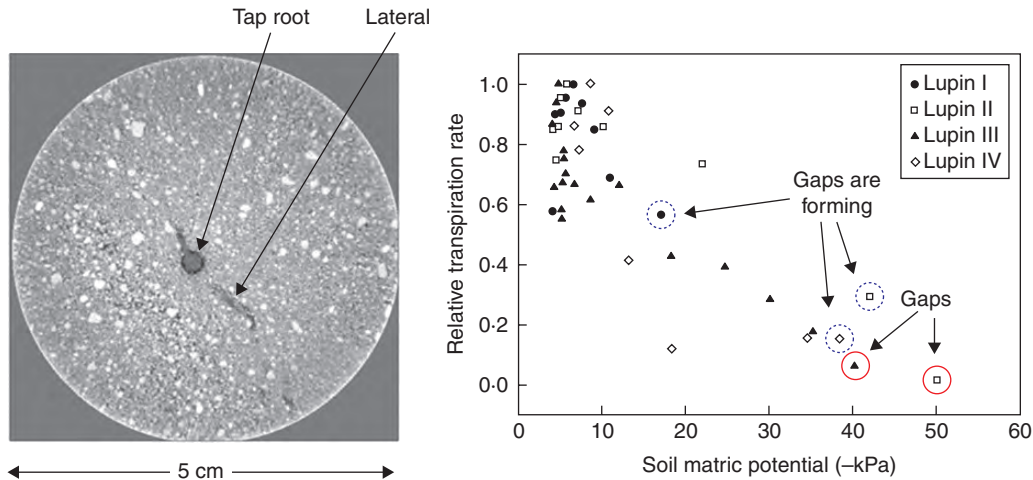


FIG. 2. Left: horizontal cross-section at 5 cm depth of lupin roots in a sandy soil at $\theta = 0.03$. An air-filled gap (black) is visible around the tap root. Right: relationship between transpiration rate, soil matric potential and gap formation for four replicates. The relative transpiration on the y-axis is the actual transpiration divided by the maximum transpiration rate of the lupins when the soil was wet. The points when gaps initiated and when continuous gaps were established are indicated with dotted and solid circles, respectively.

(Carminati *et al.*, 2010) showed that, indeed, the water content in the rhizosphere was not constantly higher or lower than that in the bulk soil, but it varied with time depending on the drying/wetting history of the soil. Neutron radiography is an imaging technique that, compared with X-ray CT, is very sensitive to water, allowing *in situ* accurate quantification of soil moisture distribution around roots of transpiring plants. Carminati *et al.* (2010) showed that water content in the rhizosphere of a large number of laterals was higher than in the bulk soil during a period of active transpiration. However, after the soil was dried to a water content of $\theta = 0.03$ and the samples were subsequently irrigated, the rhizosphere remained markedly drier than the bulk soil. It took approx. 2 d for the rhizosphere to become wet again. Neutron radiographs of a similar experiment are shown in Fig. 3. The figure shows the root and water distribution in the upper 5 cm of one of the samples. The sample was 30 cm high, 15 cm wide, and 1 cm thick, it was filled with a sandy soil, and it was planted with a lupin. The lupin has been grown for 14 d with a water table at the bottom of the sample. After 14 d, the water reservoir at the bottom of the sample was removed and the sample was left to dry for 5 d. Then the samples were wetted again from the bottom. The upper image of Fig. 3 shows the sample on day 3 of the drying period. The radiograph was processed and normalized according to Carminati *et al.* (2010) and it shows the volumetric water content in soil. In the radiograph, it can be seen that the region near the roots was wetter than the adjacent bulk soil. The lower image of Fig. 3 shows the sample on day 5, 1 h after the sample was rewetted by placing the water table at the bottom of the sample. The radiograph shows that although the bulk soil was now wet, the rhizosphere remained markedly dry. The rhizosphere became wet again after 1–2 d.

The observations reported above can only be explained if the common assumption that soil around roots has the same, static properties as bulk soil is dismissed. One factor that can explain these observations is mucilage. Mucilage, or

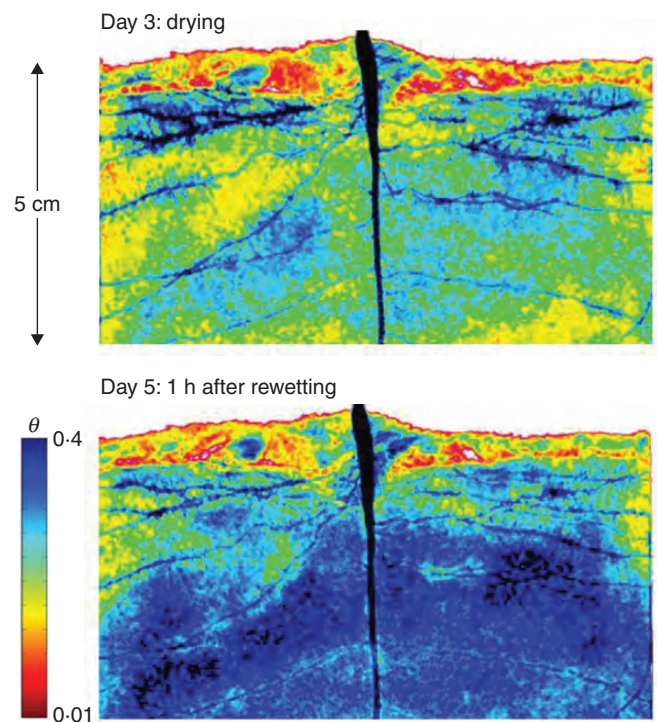


FIG. 3. Neutron radiography of water distribution around the roots of a lupin in a sandy soil. Only the upper 5 cm of the sample are shown. The upper image was taken during the drying cycle and shows a higher water content in the rhizosphere than in the bulk soil. The lower image shows the water content distribution after irrigation by capillary rise. A markedly dry rhizosphere is clearly visible. The key shows the soil water content.

mucigel, is a viscous, gluey substance produced by plants and micro-organisms. Mucilage can hold large volumes of water. At a soil matric potential of $\psi = -100$ hPa, mucilage holds an amount of water equal to 50 times its dry weight (McCully and Boyer, 1997). At this soil matric potential,

assuming a mucilage concentration in the rhizosphere of 0.1 % (dry mass of mucilage per soil weight), mucilage would increase the gravimetric water content in the rhizosphere by 5 % (weight of water per soil weight). However, the presence of phospholipids in mucilage may decrease the water-holding capacity of the rhizosphere (Read *et al.*, 2003). Mucilage therefore appears to be a good candidate to explain both the increased water content in the rhizosphere and the temporarily reduced water content after irrigation. A comprehensive review of mucilage characteristics is given below.

RHIZOSPHERE PLASTICITY: DYNAMIC ALTERATION OF THE BIMODAL DISTRIBUTION OF RHIZOSPHERE HYDRAULIC PROPERTIES

The presence of (1) air-filled gaps, (2) increased water content in the rhizosphere due to hydrated mucilage and (3) a temporarily hydrophobic rhizosphere should not be seen as contradictory characteristics but as an expression of rhizosphere plasticity. Such a plasticity may be a strategy of plants to control, for example by the rate of mucilage production, which part of the root system will have a facilitated access to water and which roots will be disconnected from the soil, for instance by air-filled gaps or by rhizosphere hydrophobicity. To describe such a dualism in rhizosphere hydraulic properties, we classify the rhizosphere into two categories: class A refers to a rhizosphere covered with hydrated mucilage. A rhizosphere of class A is expected to connect roots to soil optimally and to facilitate water flow to roots as the soil dries, as shown in Carminati *et al.* (2011). Class B refers to the case of air-filled gaps and/or a hydrophobic rhizosphere. Such a rhizosphere will partly disconnect roots from soil. Such a disconnection may be temporary and the hydraulic contact can be partially recovered according to the drying/wetting soil history. Rhizosphere of class B can avoid water loss from roots to soil and is expected to cover roots predominantly responsible for long-distance transport.

The share of root segments belonging to class A or B is highly dynamic as mucilage is primarily produced at the root tips and its production rate depends on growing conditions. Spread of mucilage in soil by diffusion depends on soil properties, soil water potential and the rate of microbial degradation (Watt *et al.*, 2006). As a result, the amount of mucilage around an individual root is variable and it depends on plants and growing conditions. Our hypothesis is that mucilage production could be used as an adaption strategy to optimize resource acquisition. Following this idea, rhizosphere hydraulic properties would show a bimodal distribution which should be accounted for in studies on root water uptake and in modelling approaches.

Our suggestion of a bimodal distribution of rhizosphere hydraulic properties is partly an analogy to the observations of McCully and co-workers (Vermeer and McCully, 1982; Wang *et al.*, 1991; Watt *et al.*, 1994), who observed bimorphic behaviour for grass roots, in particular corn roots. They distinguished between sheathed roots and bare roots. The formation of sheaths around roots (rhizosheaths) is closely linked to the presence of mucilage, and sheathed roots are reported to show higher water content compared with bare roots. As

roots develop, rhizosheath develops around young roots about 2 cm behind the tips, and sheath is lost as root age increases. Loss of soil sheath coincides with the maturation of late metaxylem vessels (Wang *et al.*, 1991).

In our concept of a bimodal distribution of rhizosphere hydraulic properties, the presence of mucilage is delaying the onset of steep gradients and thus root water uptake is maintained for extended time periods. In the concept of McCully and co-workers, the presence of mucilage coincides with high axial resistance and thus they deduce lower water uptake for these regions and regard them as partly isolated from transpiration dynamics.

Both concepts just reflect part of the picture and need to be extended to understand root water uptake at the whole plant level. Draye *et al.* (2010) recently reviewed the relative importance of soil conductivity vs. radial and axial root conductivity for root water uptake at the whole plant scale. Within a whole root system, which consists of individual roots which should be viewed as a series of segments of increasing age and developmental stage, the combination of root architecture and the ratio of radial to axial conductivity will essentially determine the long-distance propagation of the transpiration-driven negative xylem tension until the root surface, i.e. the plant's contribution to the uptake driving force at the soil–root interface (Draye *et al.*, 2010). Only when soil hydraulic conductivity is lower than most root radial conductivities does soil become limiting for uptake – this is the point where our concept of mucilage should make a difference. To cite Passioura (1980), this should be the case at intermediate and low soil water contents as ‘When the soil is wet it has little influence on the uptake of water from it to the plant [as soil hydraulic conductivity is higher than root hydraulic conductivity]. When it is dry it has a large influence [as soil hydraulic conductivity decreases exponentially as soil dries]. When it is neither wet nor dry, the extent of influence is a matter of controversy’.

Hence whether or not mucilage will make a difference for water uptake at the whole plant scale will depend on the following.

(a) Factors that affect the extent of matric potential gradient around an individual root (Gardner, 1960; Raats, 2007; Metselaar and van Lier, 2011).

Soil water content and its spatial distribution
Soil texture and related water retention curve
Transpirational demand and hence water flux per unit root surface (Carminati *et al.*, 2011)
Root diameter (i.e. length of pathway and surface area)
Inter-root distance (i.e. the length of the radial pathway for water transport).

(b) Factors related to the ratio of radial and axial root hydraulic conductivity and its spatial distribution within the root system (Draye *et al.*, 2010).

Number, size and developmental state of xylem vessels (relates to axial conductivity)
Development of apoplastic barriers such as casparian bands and suberin lamellae in the endodermis and exodermis

(relates to apoplastic transport and thus to radial conductivity)

Activity of aquaporins (relates to cell–cell path and thus radial conductivity)

Density of laterals as these break right through the exodermis, cortex and endodermis and are directly connected to the large metaxylem elements (McCully and Canny, 1988). They are thought to provide a low resistance pathway for water uptake (Häussling *et al.*, 1988; McCully and Canny, 1988).

(c) Factors related to the amount and quality of mucilage and any specific co-occurrence with any of the above parameters.

Production and release

Mucilage composition and potential function of individual components

Microbial degradation and alteration of mucilage

Spatial distribution of mucilage *in situ*

Interaction of mucilage and soil particles

Soil chemistry – nutrient uptake

Mucilage distribution in relation to root tissue differentiation and root growth.

In the following section, we will only discuss the points listed under (c) in more detail. More in-depth discussion and state of the art for the other parameters can be found in the literature cited in (a) and (b).

MUCILAGE COMPOSITION, DISTRIBUTION AND FUNCTION IN THE RHIZOSPHERE

Production and release

Mucilage secretion from plant roots primarily occurs from the cap cell at the root tip, although epidermal cells in the elongation and maturation zones may also contribute, and some authors suggest distinguishing between root cap mucilage and epidermal mucilage (Miki *et al.*, 1980; Ray *et al.*, 1988; Chaboud and Rougier, 1990). In the root cap, mucilage is produced by the Golgi apparatus in the so-called hypersecretory cells of the root cap (Paull and Jones, 1976; Battey and Blackbourn, 1993; Hawes *et al.*, 2000). According to Morre and Jones (1967), the formation can be divided into an active phase, exocytosis through the plasmalemma, and a passive phase, extrusion through the cell wall. The active phase involves Ca^{2+} for the fusion of vesicles with plasmalemma (Battey *et al.*, 1999). The passive phase is facilitated by progressive degradation of cell walls, triggered through increased activities of polygalacturonase and pectin methyl esterase as part of the release of root boarder cells from the root tip (Chaboud, 1983; Driouich *et al.*, 2006).

The mucilage production rate observed in hydroponic or aeroponic systems under axenic conditions ranges from 15 to 370 $\mu\text{g d. wt per plant}$ (Table 1). The production sometimes refers to sugars only, sometimes to total dry weight; the time span for incubation and plant age are highly variable. Data refer to the soluble fraction after separation of border cells, which by definition are part of the mucilage. Border cell release itself shows a very high variability (100–10 000 cells per root tip d^{-1}) depending on plant species and genotype

TABLE 1. Rates/amounts of mucilage production for different plant species

Mucilage production – root based	Species	Reference
15–20 μg total sugars per root tip per 20 h	<i>Zea mays</i>	Chaboud (1983)
60–370 μg sugar per plant per incubation period	<i>Zea mays</i>	Chaboud and Rougier (1990)
126 μg sugar per plant	<i>Glycine max</i>	Timotiwi and Sakurai (2002)
9.7 μg dry weight/root tip	<i>Zea mays</i>	Guinel and McCully (1986)
33–101 μg hydrated mucilage per root tip	<i>Oryza sativa</i>	Somasundaram <i>et al.</i> (2008)
100–140 μg polysaccharide cm^{-2} root surface	<i>Zea mays</i> (sand culture)	Floyd and Ohlrogge (1971)

(Driouich *et al.*, 2006; Somasundaram *et al.*, 2008). A significant positive regression between boarder cell release and mucilage production was reported by Somasundaram *et al.* (2008) for *Oryza sativa*. As no comparable data are available for other species and growing conditions, it is not clear whether the number of border cells could be used as a proxy for mucilage production in general. No data on mucilage production are available *in situ*, i.e. for soil-grown plants, with a water regime comparable with field conditions (no coarse quartz sand and through flow conditions).

Mucilage composition and potential function of individual components

The soluble fraction of mucilage is primarily composed of neutral and acid polysaccharides (about 94 %). Proteins account for about 6 % of the soluble fraction. In addition, small amounts of phenolic acids (Bacic *et al.*, 1986) and phospholipids (Read *et al.*, 2003) are detected. A particularly high fucose content is reported for purified slime of corn plants (20 % w/w). The fucose fraction is smaller in other grass species (rice and wheat, 8 % w/w) and dicotyledons analysed (Table 2). Concentrations of uronic acid are higher in dicotyledons (up to 24 %) than in graminaceous monocotyledons (<5 %) (Knee *et al.*, 2001; Table 2). The low concentrations in monocots are surprising as acidic sugars make up pectic polysaccharides which in the presence of Ca^{2+} are assumed to confer the gel properties to the root slime (Bacic *et al.*, 1986; Rinaudo, 1987). This has been shown for the model substance polygalacturonic acid (PGA) by Mimmo *et al.* (2003). In the presence of divalent calcium, PGA may form a gel with a porous and fibrillar structure that can hold a mass of water several times its own dry weight, as is described for mucilage (McCully and Boyer, 1997). Such a fibrillar structure has been reported for mucilage already within the vesicles before secretion (Battey *et al.*, 1999). Based on *in situ* observations, Cieurli *et al.* (1996) found that gel formation with PGA requires the presence of calcium and a certain extent of dehydration. Likewise, Watt *et al.* (1994) reported that drying of mucilage is a crucial element of soil adhesion.

Due to the total content and the ratio of arabinose to galactose, some authors (Clarke *et al.*, 1979; Chaboud, 1983; Bacic

TABLE 2. Mucilage composition for different plant species and cultivation systems

Mucilage composition	Species	Reference
Polysaccharides (fucose, galactose, glucose, arabinose), indication for presence of arabinogalactan proteins (AGPs)	<i>Zea mays</i>	Chaboud (1983)
94 % carbohydrates [fucose (20 %), arabinose, xylose, galactose, glucose and 3 % uronic acid], 6 % protein, 0.17 % phenolic acid; indication for presence of arabinogalactan proteins (AGPs)	<i>Zea mays</i>	Bacic <i>et al.</i> (1986)
97 % carbohydrates (11.5 % uronic acid; 9 % fucose; arabinose, xylose, mannose, galactose, glucose), 3 % proteins	<i>Vigna unguiculata</i>	Moody <i>et al.</i> (1988)
95.5 % carbohydrates [4 % uronic acid, fucose (3 %), arabinose, xylose, mannose, galactose, glucose], 5 % proteins	<i>Triticum aestivum</i>	Moody <i>et al.</i> (1988)
47.8 % (w/w) uronic acid in carbohydrate fraction (analyses of mucilage without fractionation; excised roots were used)	<i>Lepidium sativum</i>	Ray <i>et al.</i> (1988)
20–24 % (w/w) uronic acid; 1 % (w/w) fucose in carbohydrate fraction	<i>Glycine max</i>	Timotiwi and Sakurai (2002)
13 % (w/w) uronic acid; 0.5 % (w/w) fucose in carbohydrate fraction	<i>Pisum sativum</i>	Knee <i>et al.</i> (2001)

et al., 1986; Knee *et al.*, 2001) suggested that arabinogalactans and arabinogalactan proteins (AGPs) are present in mucilage. According to Fincher *et al.* (1983), AGPs have a high water-binding capacity and the ability to form gels; thus they may function as antidesiccants (Chaboud, 1983). Unlike anionic polysaccharides, AGPs do not require divalent calcium for the stabilization of their structure.

Another class of substances within the mucilage are phospholipids. These substances are only present in very small amounts. Phospholipids act as surfactants, lowering the surface tension of water and potentially decreasing the soil water-holding capacity (Read *et al.*, 2003). Both phospholipids and anionic polysaccharides are amphiphilic substances, i.e. the macromolecules possess hydrophilic and lipophilic regions and could thus be responsible for the water repellence observed in the rhizosphere (Czarnes *et al.*, 2000; Hallett *et al.*, 2003; Whalley *et al.*, 2005; Carminati *et al.*, 2010). Besides decreasing the surface tension, phospholipids will also increase the contact angle between the liquid and the solid phase. Moradi *et al.* (2012) measured a contact angle of approx. 90° in dry rhizosphere. However, the contact angle is not only affected by the chemical composition of the liquid phase but is also probably a function of the soil texture. Hence in coarse-textured soils, that have a low surface:volume ratio, mucilage can cover the particle surface more extensively, and the contact angle is expected to increase.

Microbial degradation and alteration of mucilage

Mucilage can serve as a carbon source for micro-organisms (Mary *et al.*, 1993; Knee *et al.*, 2001). The slime and root

border cells can influence the level of microbes that predominate in the immediate vicinity of roots (Hawes, 1990; Beniziri *et al.*, 2007). Knee *et al.* (2001) have shown that *Rhizobium leguminosarum*, *Burkholderia cepacia* and *Pseudomonas fluorescens* could grow to cell densities much higher than the control if mucilage was provided as the only source of carbon.

Studies on the mucilage degradation rate are scarce. Using ¹³C and ¹⁵N labelling in an incubation study with an orthic luvisol, Mary *et al.* (1993) found that the mineralization rate of mucilage-C was comparable with that of added glucose and higher than that for root tissue. However, for root mucilage, an initial lag phase of about 24 h was observed, while for glucose 36 % of the initial C had been mineralized within 24 h. After 11 d of incubation, cumulative mineralization for both substrates was 45 %. A lag phase was also observed by Morel *et al.* (1991) as they measured CO₂ evolution from their incubation experiments with corn mucilage. The highest mineralization rate without any lag phase was reported by Nguyen *et al.* (2008), with a half-life of only 3 d. The authors argued that there was no lag phase as the microbial community was already adapted to corn mucilage on that particular soil.

A lag phase could also result from antibiotic substances. Sobolev *et al.* (2006) identified prenylated stilbenes in an extract of peanut mucilage. Weisskopf *et al.* (2006) reported excretion of phenolic compounds and antifungal cell wall-degrading enzymes from cluster roots, which are sites with a particularly high abundance of roots tips.

Mucilage is not only degraded by microbes, but microbes also release substances, and in particular exopolysaccharides (EPS). The mixture of plant-derived slime, released root border cells and bacterial EPS is referred to as mucigel. Well known representatives of bacterial EPS are the anionic polymer xanthan (composed of glucose, mannose and pyruvic acid) and the neutral polymer dextran (composed of glucose). Xanthan shows a conformation similar to that of PGA, allowing for high water-holding capacity. Dextran, on the other hand, has a random coil structure and retains less water for a given water potential (Chenu and Roberson, 1996).

EPS can hold approximately ten times their own weight in water at a water potential of −0.1 MPa (Chenu and Roberson, 1996). This is much more than what soil may hold, and slightly less than the values reported by McCully and Boyer (1997) for mucilage.

Spatial distribution of mucilage in situ: methodological approaches

Mucilage/mucigel appears under the electron microscope as an unevenly distributed layer of granular and fibrillar material covering the outer surface of the root (Greaves and Darbyshire, 1972). The mucilage layer extends up to 50 µm from the root surface (Oades, 1978; Foster, 1988) in thin sections of embedded soil.

Different stains have been used, some of them specific to individual components of mucilage such as fucose [anti-fucose antibodies (Roy *et al.*, 2002) or lectins (Watt *et al.*, 1993)] and some of them specific to functional groups (ruthenium red, toluidine blue, alcian blue, periodic acid, sodium

borohydride, hydrolases, esterase; for details, see Watt *et al.*, 1993). Most of the methods used so far for visualizing mucilage involve addition of liquids and washing steps. However, mucilage including border cells readily diffuses away from the root surface or partly dissolves as soon as it comes into contact with water (Driouich *et al.*, 2006). Mucilage diffusion depends on its age and potential previous drying cycles (Watt *et al.*, 1994). Hence, these methods carry some risk of changing the original mucilage distribution. Unfortunately, none of these methods enables non-destructive observation *in situ*. The method introduced by Watt *et al.* (1993), involving the use of a nylon blotting membrane, has some potential to be used for *in situ* observations similar to the use of filter papers for different root exudates (Dinkelaker *et al.*, 1993). A potential alternative could also be the use of DRIFT (diffuse reflectance infrared Fourier transform) spectral analysis which has recently been used for mapping the spatial distribution of organic matter in soil (Ellerbrock *et al.*, 2009). Interestingly, this method can be applied to fresh samples without infiltration of chemicals.

Interaction between mucilage and soil particles

The effect of mucilage on the water-holding capacity of the soil is controversial. Because of the high water-holding capacity of mucilage (McCully and Boyer, 1997), it could be thought that mucilage always increases the water-holding capacity of soils. Indeed, the experiments of Chenu (1993) show that adding anionic polysaccharides (xanthan) increases the water-holding capacity of soils. The effects were smaller for neutral polysaccharides (dextran). However, as discussed above, mucilage also contains phospholipids, which may have the opposite effect on soil. For wheat mucilage, surface tension decreases with increasing total solute concentrations, and the extent of the decrease is very similar to the effect of pure lecithin based on the same concentrations (Read *et al.*, 2003). Thus it seems that for this parameter the system is controlled by the small content of phospholipids. The sugar components of mucilage, at least the neutral monomers, are instead expected to increase surface tension (Shaw, 1980, cited in Read *et al.*, 2003). A water retention curve for different amounts of mucilage in soil or mixtures of model substances (i.e. lecithin, xanthan, Ca-PGA) is currently not available.

What component of mucilage is dominating the rhizosphere water retention curve is not yet clear, and it may depend on plant species and soil conditions. Our own findings from neutron tomography (Carminati *et al.*, 2010; Moradi *et al.*, 2011) and the results of Young (1995) showing higher water content in the immediate vicinity of roots or in rhizosheath soil support that, at least for the combination of soil and plant species investigated, the components increasing the water-holding capacity (primarily anionic polysaccharides) should dominate the mucilage system.

Opposite results were found by Walley *et al.* (2005). They measured water retention curves for aggregates collected from the rhizosphere and from the bulk soil. No difference between bulk and rhizosphere was found for wheat, while water content in the rhizosphere was reduced between -6 and -50 kPa for barley and corn. Whalley *et al.* (2005) hypothesized that the reduced water-holding capacity of

rhizosphere aggregates was caused by an increased contact angle. This hypothesis is consistent with the results of Moradi *et al.* (2012) who measured contact angles close to 90° after the rhizosphere (of lupins in sandy soil) was air dry. Note that in Whalley *et al.* (2005), the aggregates were air dried before the measurements. It may be that drying alters the effect of mucilage on soil hydraulic properties. It is known that as amphiphilic substances dry, the hydrophilic regions may all be directed to the centre of a vesicle with the hydrophobic regions forming the outer surface. Upon rewetting, such substances would then show hydrophobic properties (Czarnes *et al.*, 2000; Lichner *et al.*, 2002). In model experiments with different substances, Czarnes *et al.* (2000) observed increasing water repellence with PGA addition to soil and a number of drying cycles, but not with dextran and xanthan.

If the effect of mucilage on the soil water retention curve is controversial, its effect on the hydraulic conductivity is even less clear. Because of increased viscosity and pore clogging, mucilage is expected to decrease the soil saturated conductivity by one or more orders of magnitude (Or *et al.*, 2007). The effects on unsaturated conductivity are not known. In cases where mucilage increases the soil water-holding capacity, it can be expected that the unsaturated conductivity at relatively low water potential increases. This fact can be understood by comparing fine and coarse textured soils. Coarse soils have larger pores and higher saturated conductivity. However, as they are drained more easily than fine textured soils, the latter have higher unsaturated conductivity. A different situation may occur when mucilage acts as a surfactant. Passioura (1988) speculated that by decreasing the surface tension of the soil solution, mucilage would help roots to drain otherwise unavailable water from small pores. However, such water, besides being of very small quantity, will be drained by the soil as well as by the plant, resulting in a relatively dry rhizosphere with, probably, a limited conductivity. In this case, the effect on root water uptake may be fairly negative. Further experiments are needed to measure the unsaturated conductivity of the rhizosphere.

Besides altering the hydraulic properties of the rhizosphere, mucilage also alters the structure and mechanical properties of the rhizosphere. Mucilage increases adhesive forces that stabilize soil aggregates in the rhizosphere (Morel, 1991; Watt *et al.*, 1993, 1994; Amellal *et al.*, 1999; Czarnes *et al.*, 2000). Czarnes *et al.* (2000) demonstrated higher tensile strength for treatments with PGA and xanthan, and the effect was maintained throughout several drying cycles. Watt *et al.* (1993) discussed in detail possible mechanisms involved in soil binding – their data do not support the role of anionic polymers with carboxyl groups binding to charged metal ions attached to soil microaggregates. They emphasize the importance of hydroxyl groups of terminally branched neutral sugars and suggest hydrogen-bonding mechanisms between these groups and the oxygen surfaces of soil particles. This mechanism would be less dependent on pH changes compared with anionic polymers. The observation that a drying cycle is required for stable binding is explained by the increase in contact between mucilage molecules and the soil surface as a result of partial dehydration.

Finally, we note that soil hydraulic properties with and without mucilage model components have been mostly investigated for homogenized samples, i.e. all of the soil matrix having the same properties. In reality, only a small zone of soil around roots (in the range of micrometres to millimetres) will be impregnated with the substance and it may well be that the average values measured in the collected samples underestimate the effect of mucilage in the rhizosphere, with hydraulic properties changing as a function of distance from the root.

Soil chemistry: nutrient availability and mucilage

Rhizosphere hydraulic properties and resulting root–soil contact are crucial for transport of nutrients to the root surface and for the spread of substances released by the roots to mobilize nutrients from the soil – these issues are already covered above.

However, there are also more specific interactions for individual nutrients. In the case of phosphorus (P), the link between the presence of mucilage and P uptake is not just through transport by mass flow and diffusion, which are both functions of soil water content, but also through alteration of P sorption to the soil minerals (Matar *et al.*, 1967). This can be due to the phospholipid components of the mucilage (Read *et al.*, 2003) or mediated by the galacturonate component of the polysaccharide (Nagarajah *et al.*, 1970). Gaume *et al.* (2000) observed a marked decrease of P sorption to ferrihydrite which was pre-treated with mucilage or PGA. Similar studies by Mikutta *et al.* (2006) showed only small effects on P solubility and different behaviour of mucilage and PGA. Neither of the two studies stabilized PGA by defined additions of calcium, nor did they report the calcium status of the soil.

Mucilage and the resulting change in rhizosphere hydraulic properties may also alter well known mechanisms of P mobilization via release of organic acid anions. The exudation is not uniform along roots, and some species form specific root types with extremely dense spacing of lateral rootlets and root hairs, called cluster roots under P limitation. Due to overlapping absorption zones, this local increase in root surface is not thought to serve as a large area for uptake, but rather as a large capacity to induce locally an intense change of soil chemical properties, which is required for efficient P solubilization (Neumann and Martinoia, 2002). As the large number of laterals forming a cluster root is also a hot-spot of mucilage production (Lamont, 2003), due to the enormous number of root tips within a small soil volume, a strong interaction between mucilage production, organic acid release (Deina *et al.*, 2003) and soil hydraulics can be expected. Such interactions have not been investigated so far.

For calcium, a very specific interaction can be expected as Ca^{2+} forms bridges between anionic polysaccharide chains (Cortez and Billes, 1982) and may thus determine the stereo-logical conformation and thus the physical properties of mucilage (Mimmo *et al.*, 2003). The binding of Ca^{2+} to carboxyl groups strongly depends on the concentration of competing cations which may be Al^{3+} or Cu^{2+} (Deiana *et al.*, 2003; Mimmo *et al.*, 2003). In addition, dissociation of carboxyl groups will be strongly pH dependent and thus mucilage conformation should be affected by soil pH if anionic

polysaccharides dominate the physical and chemical properties of mucilage.

Additional complexity is added by the fact that mucilage not only has an indirect effect on diffusion via changes in soil water content, but directly hinders anion diffusion due to fixed negative charges in some polysaccharides (Hart *et al.*, 2001).

Mucilage distribution in relation to root tissue differentiation and root growth

McCully and co-workers have shown that loss of rhizosheaths from corn roots coincides with maturation of late metaxylem elements in corn roots (*Zea mays* L. cv. Seneca Chief) (St Aubin *et al.*, 1986; McCully and Canny, 1988; Wang *et al.*, 1991). Immature metaxylem was observed in their studies up to 20–30 cm from the root tip. Early metaxylem is mature much closer to the tip (McCully and Canny, 1988).

These observations are in line with the results from Steudle and Peterson (1998) also for corn roots. They reported mature early metaxylem vessels at about 25 mm from the tip and mature late metaxylem vessels about 250 mm from the tip. In contrast to McCully and co-workers, they include results from root pressure probe measurements in their discussion, and conclude that as soon as early metaxylem vessels mature, the axial hydraulic resistance within the xylem is usually not rate limiting for uptake. They regard only the 25 mm with immature early metaxylem as hydraulically isolated. More data about temporal changes of root radial and axial conductivity for a range of plant species and for soil-grown plants are required to draw any conclusions regarding the relationship between the presence of mucilage and root hydraulic properties.

The distance to the root tip is often used to describe the developmental stage as it is easy to measure (Lecompte *et al.*, 2001). However, due to the large variability in growth rate ($2\text{--}69\text{ mm d}^{-1}$) depending on plant species, root type and growing conditions (temperature, salt content and soil moisture) (for a review, see tables 5 and 6 in Watt *et al.*, 2006), distance to the tip is a poor measure of root development (differentiation). Watt *et al.* (2006) discusses in detail the problem of calculating the time span for development of a rhizosphere at a certain location in relation to root development. Root age might be a much better proxy for root development and duration of contact of soil particles with a root in a particular stage.

RHIZOSPHERE AND SOIL–PLANT WATER RELATIONS

In the previous sections we reviewed the properties of the rhizosphere. Rhizosphere properties span from the case of air-filled gaps between soil and roots, to the case of hydrated mucilage around roots. In Fig. 1 we showed that as soils dry, significant gradients in water potential and water content occur in the soil near the roots. Such gradients are affected by the rhizosphere properties. The effects of varying hydraulic properties on root water uptake have been investigated in a modelling study by Carminati *et al.* (2011). These authors

assumed that the rhizosphere had a higher water-holding capacity than the bulk soil at each matric potential. This assumption was justified by the observation that during a drying period the rhizosphere was wetter than the bulk soil, even at the peak of transpiration (Carminati *et al.*, 2010). However, the assumption is not valid for the period subsequent to irrigation, when the rhizosphere was temporarily water repellent. The assumption of Carminati *et al.* (2011) refers to the case of hydrated mucilage. The more general case illustrating the rhizosphere dynamics is discussed in Carminati (2012). Carminati *et al.* (2011) assumed that the saturated conductivity of the rhizosphere was lower than that of the bulk soil, because of increased viscosity due to mucilage. At negative water potentials, the unsaturated conductivity of the rhizosphere was assumed to be higher than that of the bulk soil. The reason for this assumption is that at negative water potentials the rhizosphere remains wetter and more conductive than the bulk soil. Under these assumptions, Carminati *et al.* (2011) calculated the profile of water potentials towards a root during a drying cycle. The simulation boundary conditions were similar to those in Fig. 1. The results showed that, due to its high hydraulic conductivity at negative matric potentials, the rhizosphere reduced the drop in matric potential near roots as the soil dried. The soil could be drained for longer and to a lower water content until the root surface reached the wilting point, assumed to occur at $\psi = -1.5$ MPa. In this way, the rhizosphere favoured water availability to plants as soil dried. Despite the fact that this mechanism originates from the higher water-holding capacity of the rhizosphere, the real function of the mucilage is not to be a water capacitor, but rather to keep the root hydraulically connected to the soil. The effect of mucilage is similar to increasing the root radius.

These theoretical findings need to be confirmed by *in situ* measurements of rhizosphere conductivity. Carminati (2012) extended the model to include the fact that the rhizosphere becomes hydrophobic after severe drying. The model results show that a hydrophobic rhizosphere requires larger gradients in water potential to take up water at the same rate. In other words, the local root water uptake may be restricted by where and when rhizosphere becomes hydrophobic. Similarly, a local decrease in water uptake is expected when air-filled gaps are present between soil and roots.

Controversy remains about the role of mucilage in water-holding capacity. Does exuded mucilage increase or decrease the water-holding capacity of soils? How do the properties of mucilage change over time and during drying/wetting cycles? Mucilage contains both components acting as sponges (high water-holding capacity) and components acting as surfactants, and eventually becoming hydrophobic upon drying (reducing water-holding capacity). The relative importance of the two components seems to vary over time and is a function of the plant species and growing conditions. As explained above, our personal observations suggest that mucilage increases the water-holding capacity of the rhizosphere near young roots (plants between 2 and 4 weeks old) growing in sandy soils. However, the rhizosphere may become temporarily hydrophobic after drying, suggesting that mucilage temporarily decreases the water-holding capacity after being dry. Again, the two facts should not be seen as a

contradiction, but as an expression of the rhizosphere's plasticity and its time dynamics.

The question now is how to extend such rhizosphere plasticity to the whole root system and to evaluate the effects at the plant scale. Our idea is that the rhizosphere properties have a bimodal distribution, with rhizosphere of class A describing a 'good' hydraulic contact between soil and roots (e.g. by hydrated mucilage), and rhizosphere of class B describing a 'loose' connection between soil and roots (e.g. by gaps or hydrophobicity). Classification of the rhizosphere properties and assigning roots to class A and B depends on several factors such as mucilage production, mucilage degradation rate, root elongation, growing conditions and variation among plant species. These factors have been reviewed above and in Watt *et al.* (2006). Typically, we expect rhizosphere of class A along young and more distal roots and rhizosphere of class B along older and more proximal roots. This concept is illustrated in Fig. 4.

What are the consequences of a bimodal distribution of rhizosphere properties on root water uptake? According to McCully and co-workers, the presence of mucilage and rhizosheaths (Vermeer and McCully, 1982; Wang *et al.*, 1991; Watt *et al.*, 1994) corresponds to high axial resistance. This observation led them to conclude that these regions are partly isolated from the transpiration stream and are therefore not relevant for root water uptake. This conclusion may be justified if mucilage and rhizosheaths cover just the root tips; which may be true if root elongation and mucilage degradation are both fast, but may not be the case in general. Furthermore, local water uptake depends not only on the degree of xylem development and the relative axial hydraulic conductivity, but also on the relative importance of radial and axial root conductivity and soil conductivity (Draye *et al.*, 2010). Zwieniecki *et al.* (2003) showed that when roots have uniform and good access to water, 30 % of the root surface is able to take up 90 % of the total water uptake, and that the most apical parts are isolated from the transpiration stream. Zwieniecki *et al.* (2003) explained their results by describing the root as a porous pipe and that the local uptake depends on the ratio between radial and axial conductance. When axial conductance is relatively high, root water uptake is uniform along

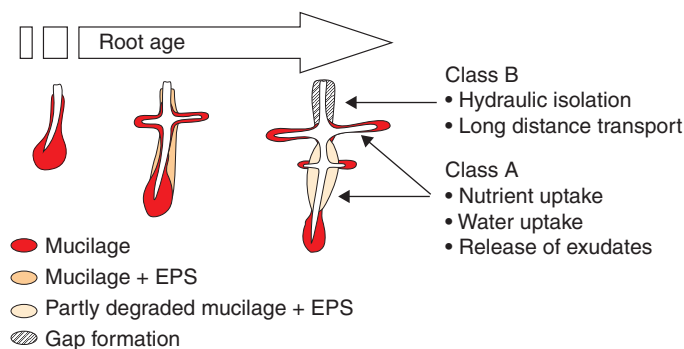


FIG. 4. Diagram of the temporal and spatial development of roots and their rhizosphere from roots belonging to class A (having favourable properties for root water uptake) to roots belonging to class B (being hydraulically isolated from the surrounding soil).

roots. When axial conductance decreases, pressure dissipation along the xylem is significant and the more distal parts do take up less water. Findings by Zwieniecki *et al.* (2003) support the latter case and seem to justify McCully's hypothesis on the limited role of mucilage in water uptake. However, the experiments of Zwieniecki *et al.* (2003) were done with roots in nutrient solution. For roots in soil, the situation is different. As roots take up water and the soil becomes dry, the location of root water uptake also depends on the soil conductivity; and, to be precise, on the rhizosphere conductivity. Including the soil component in the axial vs. radial concept of Zwieniecki *et al.* (2003) is not difficult: the radial resistance (inverse of conductance) is the sum of the root radial resistance plus the soil resistance (with its non-linearity). The consequences of adding a soil become significant when the soil dries and its conductivity decreases down to a value comparable with those of roots. At this point, the soil acts by increasing the overall radial resistance to flow and reducing the axial component relatively. The consequence is that root water uptake increases relatively in the more distal parts, despite their lower axial conductance. Additionally, the increased conductivity of rhizosphere of class A will further increase the local uptake in the more distal parts. This simplified analysis needs to be supported by numerical models describing the water flow in soil and roots, but it shows that rhizosphere of class A may indeed help roots to acquire water as soil dries.

Rhizosphere properties are expected to become significant for root water uptake when the soil dries and its hydraulic conductivity decreases. Water uptake is then expected to shift towards the roots that have a 'better' hydraulic contact with the soil (class A), while older and more proximal roots lose hydraulic contact with the soil (class B) and serve mainly for long-distance transport.

The ecological role of the class A rhizosphere in increasing water availability to plants is straightforward. What about the role of the class B rhizosphere? Is it a problem for the whole plant that some roots lose contact with the soil? North and Nobel (1997) suggested that root isolation from soil is a strategy of plants not to lose water when the soil dries and its water potential becomes more negative than that in the roots. Looking at the whole picture, with deep roots growing in a wet soil and upper roots sitting in a dry soil, actually the concept of Nobel and co-workers seems reasonable: hydraulically it is convenient for such plants to disconnect the upper roots from the dry soil. The plant will basically feel only the deeper, wet soil. On the other hand, this mechanism will limit hydraulic lift – the water redistribution from wet to dry soil via roots at night. The real ecological implication of this mechanism is therefore not clear.

CONCLUSIONS AND FUTURE RESEARCH NEEDS

The mechanistic effect of increased water-holding capacity or altered hydraulic conductivity of the rhizosphere can be addressed in modelling exercises. However, there is a lack of experimental data on rhizosphere hydraulic properties.

Classical soil physical experimental approaches need to be scaled down to the millimetre or micrometre scale and need to take into account the geometry of the soil–root interface,

i.e. a thin layer of mucilage impregnating the soil in a radial geometry. Miniaturized devices such as microsuction cups (artificial roots) and microtensiometers can be used in combination with non-invasive methods which give detailed information on water content [neutron tomography, nuclear magnetic resonance (NMR)], soil structure and root distribution (X-ray tomography). An important parameter that needs to be measured *in situ* is the local root water uptake. Zarebanadkouki *et al.* (2012) used neutron radiography to trace deuterated water transport across and along roots. Using a convection–diffusion equation, the authors were able to quantify the net water flow across the root–soil interface. This technique can be used, for instance, to measure the effect of hydrophobicity and gaps on root water uptake.

Model experiments conducted so far have used individual components of mucilage which either are present in large quantities or are expected to dominate the physical properties of mucilage. As these model substances have different functional groups and different stereological conformation, they show completely different behaviour in the environment. Hence, much more detailed information on the chemical composition of mucilage is required, and the effect of mixtures of anionic and neutral polysaccharides and surfactants such as phospholipids needs to be investigated. As different plant species and different associations of plant species and micro-organisms show differences in their mucilage/mucigel composition, working with mixtures might be the way to draw more general conclusions.

Alternatively, the importance of mucilage for rhizosphere hydraulics could be investigated by comparing mutants with and without mucilage production. However, to our knowledge only one corn mutant is available which can be easily decapped, and this mutant shows very different growth rates in loose or compacted material in contrast to the wild type (Volness *et al.*, 2010).

Visualization of mucilage distribution *in situ* is another issue which has proved to be difficult in the past, but which could provide important information about spatial distribution of mucilage. DRIFT spectroscopy has been used to visualize the distribution of organic material in earthworm cast, and also provides some information on the quality of organic matter (Ellerbrock *et al.*, 2009). As it does not require infiltration of solutes, its potential for visualizing mucilage should be investigated. Likewise, proton NMR relaxometry is capable of characterizing the physiochemical state of water in soils. Bayer *et al.* (2010) used this technique to quantify the presence and swelling of biofilms in soils, showing the potential of this technique to quantify mucilage concentration and its hydration state.

A critical issue is the integration of altered and variable rhizosphere properties at the root system scale. For 3-D root water uptake models, there is an existing approach to include variability of root axial and radial conductivity (Doussan *et al.*, 2006; Javaux *et al.*, 2008) based on root type and age derived from root architectural models.

We suggest scaling the rhizosphere, and probably also root properties, in such models based on root age. Obviously many of the parameters which affect mucilage distribution and composition in soil and related rhizosphere hydraulic properties are dependent on time. Likewise, root axial and

hydraulic conductivities and their ratio to each other change with time. Root age has been difficult to measure in the past, but recent advances in X-ray CT analysis which enable visualization of root growth have the potential to provide 3-D maps of root age for different points in time (N. Koebernick, pers. commun.).

This concept of rhizosphere plasticity and bimodal distribution of rhizosphere properties has been derived based on our experiments with lupins in sandy soil. The situation may be different in other soil types, plant species and growing conditions. However, we think that our concept is general and can be applied to other soil–plant systems. The results reported in the literature seem to confirm this. How rhizosphere varies among (and along) soil–plant systems and to what extent it affects acquisition of water and nutrients when the resources are scarce remains an open question. However, the concepts and tools to answer this question are starting to become available.

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